Provenance of the Current-Flow Material Library

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The library

`condlib_catalog.dat` is a current-flow material library for FEMM, 258 entries in 24 folders, built 1 October 2026. A current-flow problem is solved at one frequency f, and each material is a conductivity σ (S/m), a relative permittivity ε_r and a loss tangent $\tan\delta$, combined as $\sigma_{\text{eff}} = \sigma + j\omega\varepsilon_0\varepsilon_r e^{-j\phi}$ with $\tan\phi = \tan\delta$. FEMM takes ε_r as the *magnitude* of the complex relative permittivity, so the library holds $\varepsilon'\sqrt{1+\tan^2\delta}$ where a source gives ε'; for all but a few lossy entries the two agree to better than 0.3%.

Dielectrics have one entry per tabulated frequency (60 Hz, 1 kHz, 1 MHz), named accordingly; metals, water and ground are frequency-independent over the range FEMM's current-flow solver is used for. Where a source gives a range, the entry uses its middle and the note says so; where it gives a bound, the bound is used. Leakage conductivity of insulators is from volume resistivity where one exists, otherwise zero.

The library is generated by `condlib/build_condlib.py` from one CSV file per source; `condlib/catalog.csv` lists every entry, and the table below is generated from it. `SOURCES.md` records each source document with its SHA-256 hash.

Liquids

Unlike the electrostatic and heat-flow libraries, this library needs no caution on its liquid entries. The current-flow solver models conduction and displacement current together, so water, sea water and the insulating liquids are represented correctly at any frequency their entries cover. The electrostatic library's water folder points users here for that reason.

Entries

Entry	σ (S/m)	ε_r	$\tan\delta$	Src.	Note
Metals and Alloys / Pure Metals					
Copper (IACS standard)	5.8×10^7	1	0	[9]	annealed copper standard, 100% IACS by definition
Aluminum, 99.99%	3.767×10^7	1	0	[8]	'Aluminum, 99.99%' 64.94% IACS, 2.655e-8 ohm-m
Metals and Alloys / Aluminum Alloys					
Aluminum, 1100 (O)	3.448×10^7	1	0	[1]	p. 55
Aluminum, 2024 (O)	2.941×10^7	1	0	[1]	p. 55
Aluminum, 3003 (O)	2.941×10^7	1	0	[1]	p. 55
Aluminum, 4032 (O)	2.326×10^7	1	0	[1]	p. 55
Aluminum, 5052 (all)	2.041×10^7	1	0	[1]	p. 55
Aluminum, 5056 (O)	1.695×10^7	1	0	[1]	p. 55
Aluminum, 6061 (T6)	2.5×10^7	1	0	[1]	p. 55
Aluminum, 6101 (T6)	3.333×10^7	1	0	[1]	p. 55
Aluminum, 7075 (O)	2.632×10^7	1	0	[1]	p. 55
Aluminum, 7075 (T6)	1.923×10^7	1	0	[1]	p. 55
Metals and Alloys / Copper and Copper Alloys					
Copper, pure	5.988×10^7	1	0	[1]	p. 55

Entry	**σ (S/m)**	ε_r	$\tan\delta$	**Src.**	**Note**
Electrolytic tough pitch copper (ETP)	5.848×10^7	1	0	[1]	p. 55
Oxygen-free copper (OF)	5.848×10^7	1	0	[1]	p. 55
Free-machining copper, 0.5% Te	5.495×10^7	1	0	[1]	p. 56
Free-machining copper, 1.0% Pb	5.682×10^7	1	0	[1]	p. 56
Cartridge brass, 70%	1.613×10^7	1	0	[1]	p. 56
Yellow brass	1.562×10^7	1	0	[1]	p. 56
Leaded commercial bronze	2.439×10^7	1	0	[1]	p. 56
Phosphor bronze, 1.25%	2.778×10^7	1	0	[1]	p. 56
Nickel silver, 55-18	3.226×10^6	1	0	[1]	p. 56
Low-silicon bronze (B)	6.993×10^6	1	0	[1]	p. 56
Beryllium copper	1.481×10^7	1	0	[1]	p. 56; resistivity 5.7-7.8 $\mu\Omega$-cm (middle used)
Cast Chromium copper (1% Cr)	4.762×10^7	1	0	[1]	p. 56
Cast 88Cu-8Sn-4Zn	6.667×10^6	1	0	[1]	p. 56
Cast 87Cu-10Sn-1Pb-2Zn	6.667×10^6	1	0	[1]	p. 56
Contact copper, 0.04 oxide	5.814×10^7	1	0	[1]	p. 56
Contact copper, 1.25 Sn + P	2.778×10^7	1	0	[1]	p. 56
Contact copper, 5 Sn + P	9.091×10^6	1	0	[1]	p. 56
Contact copper, 8 Sn + P	7.692×10^6	1	0	[1]	p. 56
Contact copper, 15 Zn	2.128×10^7	1	0	[1]	p. 56
Contact copper, 20 Zn	1.852×10^7	1	0	[1]	p. 57
Contact copper, 35 Zn	1.562×10^7	1	0	[1]	p. 57
Contact copper, 2 Be + Ni or Co	9.479×10^6	1	0	[1]	p. 57; resistivity 9.6-11.5 $\mu\Omega$-cm (middle used)
Metals and Alloys / Silver Alloys					
Silver, fine	6.289×10^7	1	0	[1]	p. 57
92.5Ag-7.5Cu	5×10^7	1	0	[1]	p. 57
90Ag-10Cu	5×10^7	1	0	[1]	p. 57
72Ag-28Cu	5×10^7	1	0	[1]	p. 57
72Ag-26Cu-2Ni	3.448×10^7	1	0	[1]	p. 57
85Ag-15Cd	2.028×10^7	1	0	[1]	p. 57
97Ag-3Pt	2.857×10^7	1	0	[1]	p. 57
97Ag-3Pd	3.448×10^7	1	0	[1]	p. 57
90Ag-10Pd	1.887×10^7	1	0	[1]	p. 57
90Ag-10Au	2.381×10^7	1	0	[1]	p. 57
60Ag-40Pd	4.348×10^6	1	0	[1]	p. 57
70Ag-30Pd	6.993×10^6	1	0	[1]	p. 57
Metals and Alloys / Platinum and Palladium Alloys					
Platinum	9.434×10^6	1	0	[1]	p. 57
95Pt-5Ir	5.263×10^6	1	0	[1]	p. 57
90Pt-10Ir	4×10^6	1	0	[1]	p. 57
85Pt-15Ir	3.509×10^6	1	0	[1]	p. 57
80Pt-20Ir	3.226×10^6	1	0	[1]	p. 58
75Pt-25Ir	3.03×10^6	1	0	[1]	p. 58
70Pt-30Ir	2.857×10^6	1	0	[1]	p. 58
65Pt-35Ir	2.778×10^6	1	0	[1]	p. 58
95Pt-5Ru	3.175×10^6	1	0	[1]	p. 58
90Pt-10Ru	2.326×10^6	1	0	[1]	p. 58
89Pt-11Ru	2.326×10^6	1	0	[1]	p. 58
86Pt-14Ru	2.174×10^6	1	0	[1]	p. 58
96Pt-4W	2.778×10^6	1	0	[1]	p. 58
Palladium	9.259×10^6	1	0	[1]	p. 58
95.5Pd-4.5Ru	4.132×10^6	1	0	[1]	p. 58
90Pd-10Ru	3.704×10^6	1	0	[1]	p. 58

Entry	σ (S/m)	ε_r	$\tan\delta$	Src.	Note
70Pd-30Ag	2.5×10^6	1	0	[1]	p. 58
60Pd-40Ag	2.326×10^6	1	0	[1]	p. 58
50Pd-50Ag	3.175×10^6	1	0	[1]	p. 58
72Pd-26Ag-1.71-2Ni	2.326×10^6	1	0	[1]	p. 58
60Pd-40Cu	2.857×10^6	1	0	[1]	p. 58
45Pd-30Ag-20Au-5Pt	2.564×10^6	1	0	[1]	p. 58
35Pd-30Ag-14Cu-10Pt-10Au-1Zn	2.857×10^6	1	0	[1]	p. 58
Metals and Alloys / Gold Alloys					
Gold	4.255×10^7	1	0	[1]	p. 59
90Au-10Cu	9.259×10^6	1	0	[1]	p. 59
75Au-25Ag	9.259×10^6	1	0	[1]	p. 59
72.5Au-14Cu-8.5Pt-4Ag-1Zn	5.882×10^6	1	0	[1]	p. 59
69Au-25Ag-6Pt	6.667×10^6	1	0	[1]	p. 59
41.7Au-32.5Cu-18.8Ni-7Zn	2.564×10^6	1	0	[1]	p. 59
Metals and Alloys / Resistance and Heating Alloys					
78.5Ni-20Cr-1.5Si (80-20)	9.255×10^5	1	0	[1]	p. 59
73.5Ni-20Cr-5Al-1.5Si	7.248×10^5	1	0	[1]	p. 59
68Ni-20Cr-8.5Fe-2Si	8.594×10^5	1	0	[1]	p. 59
60Ni-16Cr-22.5Fe-1.5Si	8.913×10^5	1	0	[1]	p. 59
35Ni-20Cr-43.5Fe-1.5Si	9.862×10^5	1	0	[1]	p. 59
72Fe-23Cr-5Al	7.205×10^5	1	0	[1]	p. 59
55Fe-37.5Cr-7.5Al	6.016×10^5	1	0	[1]	p. 59
Metals and Alloys / Pure Metals					
Molybdenum	1.923×10^7	1	0	[1]	p. 59
Tantalum	8.032×10^6	1	0	[1]	p. 59
Tungsten	1.77×10^7	1	0	[1]	p. 59
Metals and Alloys / Resistance and Heating Alloys					
98Cu-2Ni	2.004×10^7	1	0	[1]	p. 60
94Cu-6Ni	1.007×10^7	1	0	[1]	p. 60
89Cu-11Ni	6.684×10^6	1	0	[1]	p. 60
78Cu-22Ni	3.342×10^6	1	0	[1]	p. 60
55Cu-45Ni (constantan)	2.005×10^6	1	0	[1]	p. 60
87Cu-13Mn (manganin)	2.074×10^6	1	0	[1]	p. 60
83Cu-13Mn-4Ni (manganin)	2.074×10^6	1	0	[1]	p. 60
85Cu-10Mn-4Ni (shunt manganin)	2.616×10^6	1	0	[1]	p. 60
70Cu-20Ni-10Mn	2.046×10^6	1	0	[1]	p. 60
67Cu-5Ni-27Mn	1.003×10^6	1	0	[1]	p. 60
Metals and Alloys / Pure Metals					
Nickel, 99.8%	1.253×10^7	1	0	[1]	p. 60
Metals and Alloys / Resistance and Heating Alloys					
71Ni-29Fe	5.013×10^6	1	0	[1]	p. 60
80Ni-20Cr	8.913×10^5	1	0	[1]	p. 60
75Ni-20Cr-3Al + Cu or Fe	7.52×10^5	1	0	[1]	p. 61
76Ni-17Cr-4Si-3Mn	7.52×10^5	1	0	[1]	p. 61
60Ni-16Cr-24Fe	8.913×10^5	1	0	[1]	p. 61
35Ni-20Cr-45Fe	9.862×10^5	1	0	[1]	p. 61
72Fe-23Cr-5Al-0.5Co	7.381×10^5	1	0	[1]	p. 61
Metals and Alloys / Pure Metals					
Iron, 99.99%	1.03×10^7	1	0	[1]	p. 61
Metals and Alloys / Steels and Irons					
Low-carbon iron (relay, annealed)	1×10^7	1	0	[1]	p. 64
Steel, 1010 (relay, annealed)	8.333×10^6	1	0	[1]	p. 64
Silicon steel, 1% Si	4.348×10^6	1	0	[1]	p. 64
Silicon steel, 2.5% Si	2.439×10^6	1	0	[1]	p. 64
Silicon steel, 3% Si	2.083×10^6	1	0	[1]	p. 64

Entry	σ (S/m)	ε_r	$\tan\delta$	Src.	Note
Silicon steel, 3% Si, grain-oriented	2.083×10^6	1	0	[1]	p. 64
Silicon steel, 4% Si	1.695×10^6	1	0	[1]	p. 64
Metals and Alloys / Stainless Steels					
Stainless, Type 410	1.754×10^6	1	0	[1]	p. 64
Stainless, Type 416	1.754×10^6	1	0	[1]	p. 64
Stainless, Type 430	1.667×10^6	1	0	[1]	p. 64
Stainless, Type 443	1.471×10^6	1	0	[1]	p. 65
Stainless, Type 446	1.639×10^6	1	0	[1]	p. 65
Stainless, Type 302	1.389×10^6	1	0	[1]	p. 65
Stainless, Type 309	1.282×10^6	1	0	[1]	p. 65
Stainless, Type 316	1.351×10^6	1	0	[1]	p. 65
Stainless, Type 317	1.351×10^6	1	0	[1]	p. 65
Stainless, Type 347	1.37×10^6	1	0	[1]	p. 65
Stainless, Type 403	1.754×10^6	1	0	[1]	p. 65
Stainless, Type 405	1.667×10^6	1	0	[1]	p. 65
Stainless, Type 501	2.5×10^6	1	0	[1]	p. 65
Cast heat-resistant alloy HH	1.25×10^6	1	0	[1]	p. 65
Cast heat-resistant alloy HK	1.111×10^6	1	0	[1]	p. 65
Cast heat-resistant alloy HT	1×10^6	1	0	[1]	p. 65
Metals and Alloys / Steels and Irons					
Steel, A36	6.25×10^6	1	0	[2]	Callister Table B.9: at 50-100 C
Steel, 1020 (annealed)	6.25×10^6	1	0	[2]	Callister Table B.9: at 50-100 C
Steel, 1040 (annealed)	6.25×10^6	1	0	[2]	Callister Table B.9: at 50-100 C
Steel, 4140 (quenched and tempered)	4.545×10^6	1	0	[2]	
Steel, 4340 (quenched and tempered)	4.032×10^6	1	0	[2]	
Metals and Alloys / Stainless Steels					
Stainless, 304 (annealed)	1.389×10^6	1	0	[2]	
Stainless, 440A (annealed)	1.667×10^6	1	0	[2]	
Stainless, 17-4PH (annealed)	1.02×10^6	1	0	[2]	
Metals and Alloys / Cast Irons					
Gray iron, G1800	6.667×10^5	1	0	[2]	
Gray iron, G3000	1.053×10^6	1	0	[2]	
Gray iron, G4000	1.176×10^6	1	0	[2]	
Ductile iron, 60-40-18	1.818×10^6	1	0	[2]	
Ductile iron, 80-55-06	1.613×10^6	1	0	[2]	
Ductile iron, 120-90-02	1.613×10^6	1	0	[2]	
Metals and Alloys / Aluminum Alloys					
Aluminum, 6061 (annealed)	2.703×10^7	1	0	[2]	
Aluminum, 356.0 (T6)	2.262×10^7	1	0	[2]	
Metals and Alloys / Copper and Copper Alloys					
Copper, C36000 (free-cutting brass)	1.515×10^7	1	0	[2]	
Copper, C71500 (30% copper-nickel)	2.667×10^6	1	0	[2]	
Copper, C93200 (bearing bronze)	6.944×10^6	1	0	[2]	
Metals and Alloys / Magnesium Alloys					
Magnesium, AZ31B	1.087×10^7	1	0	[2]	
Magnesium, AZ91D	5.882×10^6	1	0	[2]	
Metals and Alloys / Titanium Alloys					
Titanium, CP Grade 1	2.128×10^6	1	0	[2]	Callister Table B.9: 4.2-5.2 x 10^{-7}; middle of range
Titanium, Ti-5Al-2.5Sn	6.369×10^5	1	0	[2]	
Titanium, Ti-6Al-4V	5.848×10^5	1	0	[2]	
Metals and Alloys / Nickel and Cobalt Alloys					
Nickel 200	1.053×10^7	1	0	[2]	
Inconel 625	7.752×10^5	1	0	[2]	
Monel 400	1.828×10^6	1	0	[2]	

Entry	σ (S/m)	ε_r	$\tan\delta$	Src.	Note
Haynes 25	1.124×10^6	1	0	[2]	
Invar	1.22×10^6	1	0	[2]	
Super Invar	1.25×10^6	1	0	[2]	
Kovar	2.041×10^6	1	0	[2]	
Metals and Alloys / Other Metals					
Lead, chemical	4.854×10^6	1	0	[2]	
Lead, antimonial (6% Sb)	3.953×10^6	1	0	[2]	
Tin, commercially pure	9.009×10^6	1	0	[2]	
Solder, 60Sn-40Pb	6.667×10^6	1	0	[2]	
Zirconium, reactor grade 702	2.519×10^6	1	0	[2]	
Nonmetallic Conductors					
Silicon carbide heating element	6.667×10^5	1	0	[1]	resistivity 100-200 $\mu\Omega$-cm (middle used)
Molybdenum disilicide (MoSi2)	2.685×10^6	1	0	[1]	
Graphite (heating element)	1.099×10^5	1	0	[1]	
Graphite, extruded (with grain)	7.407e+04	1	0	[2]	7-20 x 10^{-6} ohm-m (middle used)
Graphite, isostatically molded	7.143e+04	1	0	[2]	10-18 x 10^{-6} ohm-m (middle used)
Solid Insulation / 60 Hz					
Alumina (Al2O3) (60 Hz)	1×10^{-13}	8.5	0.001	[3]	rho: Callister B.9 aluminum oxide 99.9% $>10^{13}$ ohm-m (bound)
Solid Insulation / 1 kHz					
Alumina (Al2O3) (1 kHz)	1×10^{-13}	8.5	0.001	[3]	rho: Callister B.9 aluminum oxide 99.9% $>10^{13}$ ohm-m (bound)
Solid Insulation / 1 MHz					
Alumina (Al2O3) (1 MHz)	1×10^{-13}	8.5	0.001	[3]	rho: Callister B.9 aluminum oxide 99.9% $>10^{13}$ ohm-m (bound)
Solid Insulation / 60 Hz					
Porcelain (mullite) (60 Hz)	0	8.2	0.0014	[3]	
Solid Insulation / 1 kHz					
Porcelain (mullite) (1 kHz)	0	8.2	5.7×10^{-4}	[3]	
Solid Insulation / 1 MHz					
Porcelain (mullite) (1 MHz)	0	8.2	2×10^{-4}	[3]	
Solid Insulation / 60 Hz					
Steatite (60 Hz)	0	5.5	0.0013	[3]	
Solid Insulation / 1 kHz					
Steatite (1 kHz)	0	5	4.5×10^{-4}	[3]	
Solid Insulation / 1 MHz					
Steatite (1 MHz)	0	5	3.7×10^{-4}	[3]	
Solid Insulation / 60 Hz					
Magnesium oxide (MgO) (60 Hz)	0	9.65	3×10^{-4}	[3]	dissipation factor <3 x 10^{-4} (bound used)
Solid Insulation / 1 kHz					
Magnesium oxide (MgO) (1 kHz)	0	9.65	3×10^{-4}	[3]	dissipation factor <3 x 10^{-4} (bound used)
Solid Insulation / 1 MHz					
Magnesium oxide (MgO) (1 MHz)	0	9.69	3×10^{-4}	[3]	dissipation factor <3 x 10^{-4} (bound used)
Solid Insulation / 60 Hz					
Glass (soda lime) (60 Hz)	3.165×10^{-11}	6.25	0.005	[3]	rho: Callister B.9 10^{10}-10^{11} ohm-m (geometric middle)
Solid Insulation / 1 kHz					
Glass (soda lime) (1 kHz)	3.165×10^{-11}	6.16	0.0042	[3]	rho: Callister B.9 10^{10}-10^{11} ohm-m (geometric middle)
Solid Insulation / 1 MHz					
Glass (soda lime) (1 MHz)	3.165×10^{-11}	6	0.0027	[3]	rho: Callister B.9 10^{10}-10^{11} ohm-m (geometric middle)

Entry	σ (S/m)	ε_r	$\tan\delta$	Src.	Note
Solid Insulation / 60 Hz					
Mica (60 Hz)	0	6.9	0.0015	[3]	
Solid Insulation / 1 kHz					
Mica (1 kHz)	0	6.9	2×10^{-4}	[3]	
Solid Insulation / 1 MHz					
Mica (1 MHz)	0	5.4	3.5×10^{-4}	[3]	
Solid Insulation / 1 kHz					
SiO2 film (1 kHz)	1×10^{-18}	3.9	7×10^{-4}	[3]	rho: Callister B.9 fused silica $>10^{18}$ ohm-m (bound)
Si3N4 (1 kHz)	1×10^{-12}	12.7	1×10^{-4}	[3]	dissipation factor <1 x 10^{-4} (bound used); rho: Callister B.9 silicon nitride $>10^{12}$ ohm-m (bound)
Ta2O5 (1 kHz)	0	28	0.01	[3]	
HfO2 (1 kHz)	0	35	0.01	[3]	
Solid Insulation / 60 Hz					
Polyethylene (LDPE) (60 Hz)	1.414×10^{-16}	2.3	2×10^{-4}	[3]	rho: Callister B.9 10^{15}-5x10^{16} ohm-m (geometric middle)
Solid Insulation / 1 kHz					
Polyethylene (LDPE) (1 kHz)	1.414×10^{-16}	2.3	2×10^{-4}	[3]	rho: Callister B.9 10^{15}-5x10^{16} ohm-m (geometric middle)
Solid Insulation / 1 MHz					
Polyethylene (LDPE) (1 MHz)	1.414×10^{-16}	2.3	2×10^{-4}	[3]	rho: Callister B.9 10^{15}-5x10^{16} ohm-m (geometric middle)
Solid Insulation / 60 Hz					
Polyethylene (MDPE) (60 Hz)	1.414×10^{-16}	2.3	2×10^{-4}	[3]	rho: Callister B.9 LDPE/HDPE range 10^{15}-5x10^{16} ohm-m (geometric middle)
Solid Insulation / 1 kHz					
Polyethylene (MDPE) (1 kHz)	1.414×10^{-16}	2.3	2×10^{-4}	[3]	rho: Callister B.9 LDPE/HDPE range 10^{15}-5x10^{16} ohm-m (geometric middle)
Solid Insulation / 1 MHz					
Polyethylene (MDPE) (1 MHz)	1.414×10^{-16}	2.3	2×10^{-4}	[3]	rho: Callister B.9 LDPE/HDPE range 10^{15}-5x10^{16} ohm-m (geometric middle)
Solid Insulation / 60 Hz					
Polyethylene (HDPE) (60 Hz)	1.414×10^{-16}	2.35	2×10^{-4}	[3]	rho: Callister B.9 10^{15}-5x10^{16} ohm-m (geometric middle)
Solid Insulation / 1 kHz					
Polyethylene (HDPE) (1 kHz)	1.414×10^{-16}	2.35	2×10^{-4}	[3]	rho: Callister B.9 10^{15}-5x10^{16} ohm-m (geometric middle)
Solid Insulation / 1 MHz					
Polyethylene (HDPE) (1 MHz)	1.414×10^{-16}	2.35	2×10^{-4}	[3]	rho: Callister B.9 10^{15}-5x10^{16} ohm-m (geometric middle)
Solid Insulation / 60 Hz					
Cross-linked polyethylene (XLPE) (60 Hz)	0	2.3	3×10^{-4}	[3]	
Solid Insulation / 1 MHz					
Cross-linked polyethylene (XLPE) (1 MHz)	0	2.28	4×10^{-4}	[3]	
Solid Insulation / 60 Hz					
EPR (60 Hz)	0	3.25	0.004	[3]	dielectric constant 3.0-3.5 given at 1 kHz only (middle used); dissipation factor at 60 Hz

Entry	σ (S/m)	ε_r	$\tan\delta$	Src.	Note
Polypropylene (60 Hz)	1×10^{-14}	2.25	2.5×10^{-4}	[3]	ranges 2.22-2.28 / 2-3 x 10^{-4} / 2.5-3.0 x 10^{-4} (middles used); rho: Callister B.9 $>10^{14}$ ohm-m (bound)
Solid Insulation / 1 kHz					
Polypropylene (1 kHz)	1×10^{-14}	2.25	2.75×10^{-4}	[3]	ranges 2.22-2.28 / 2-3 x 10^{-4} / 2.5-3.0 x 10^{-4} (middles used); rho: Callister B.9 $>10^{14}$ ohm-m (bound)
Solid Insulation / 1 MHz					
Polypropylene (1 MHz)	1×10^{-14}	2.25	4.6×10^{-4}	[3]	ranges 2.22-2.28 / 2-3 x 10^{-4} / 2.5-3.0 x 10^{-4} (middles used); rho: Callister B.9 $>10^{14}$ ohm-m (bound)
Solid Insulation / 60 Hz					
PTFE (60 Hz)	1×10^{-17}	2	2×10^{-4}	[3]	dissipation factor <2 x 10^{-4} (bound used); rho: Callister B.9 10^{17} ohm-m
Solid Insulation / 1 kHz					
PTFE (1 kHz)	1×10^{-17}	2	2×10^{-4}	[3]	dissipation factor <2 x 10^{-4} (bound used); rho: Callister B.9 10^{17} ohm-m
Solid Insulation / 1 MHz					
PTFE (1 MHz)	1×10^{-17}	2	2×10^{-4}	[3]	dissipation factor <2 x 10^{-4} (bound used); rho: Callister B.9 10^{17} ohm-m
Solid Insulation / 60 Hz					
Glass-reinforced polyester premix (60 Hz)	1×10^{-13}	6.302	0.025	[3]	ranges 5.3-7.3 / 1-4 x 10^{-2} / 5.0-6.4 / 0.8-2.2 x 10^{-2} (middles used); rho: Callister B.9 thermoset polyester 10^{13} ohm-m
Solid Insulation / 1 MHz					
Glass-reinforced polyester premix (1 MHz)	1×10^{-13}	5.701	0.015	[3]	ranges 5.3-7.3 / 1-4 x 10^{-2} / 5.0-6.4 / 0.8-2.2 x 10^{-2} (middles used); rho: Callister B.9 thermoset polyester 10^{13} ohm-m
Solid Insulation / 60 Hz					
Thermoplastic polyester (60 Hz)	0	3.55	0.00175	[3]	dielectric constant 3.3-3.8 given at 100 Hz (middle used); dissipation factor 1.5-2.0 x 10^{-3} at 60 Hz (middle used)
Polycarbonate (60 Hz)	5×10^{-15}	3.17	9×10^{-4}	[3]	rho: Callister B.9 2 x 10^{14} ohm-m
Solid Insulation / 1 MHz					
Polycarbonate (1 MHz)	5×10^{-15}	2.96	0.01	[3]	rho: Callister B.9 2 x 10^{14} ohm-m
Solid Insulation / 60 Hz					
Epoxy (mineral filler) (60 Hz)	3.165×10^{-12}	5.006	0.047	[3]	ranges 4.4-5.6 / 1.1-8.3 x 10^{-2} / 4.2-4.9 / 0.19-1.4 x 10^{-1} / 4.1-4.6 / 0.13-1.4 x 10^{-1} (middles used); rho: Callister B.9 epoxy 10^{10}-10^{13} ohm-m (geometric middle)
Solid Insulation / 1 kHz					
Epoxy (mineral filler) (1 kHz)	3.165×10^{-12}	4.565	0.08	[3]	ranges 4.4-5.6 / 1.1-8.3 x 10^{-2} / 4.2-4.9 / 0.19-1.4 x 10^{-1} / 4.1-4.6 / 0.13-1.4 x 10^{-1} (middles used); rho: Callister B.9 epoxy 10^{10}-10^{13} ohm-m (geometric middle)
Solid Insulation / 1 MHz					

Entry	σ (S/m)	ε_r	$\tan\delta$	Src.	Note
Epoxy (mineral filler) (1 MHz)	3.165×10^{-12}	4.363	0.077	[3]	ranges 4.4-5.6 / 1.1-8.3 x 10^{-2} / 4.2-4.9 / 0.19-1.4 x 10^{-1} / 4.1-4.6 / 0.13-1.4 x 10^{-1} (middles used); rho: Callister B.9 epoxy 10^{10}-10^{13} ohm-m (geometric middle)
Solid Insulation / 60 Hz					
Epoxy (silica filler) (60 Hz)	3.165×10^{-12}	3.851	0.019	[3]	ranges 3.2-4.5 / 0.8-3.0 x 10^{-2} / 3.2-4.0 / 0.8-3.0 x 10^{-2} / 3.0-3.8 / 2-4 x 10^{-2} (middles used); rho: Callister B.9 epoxy 10^{10}-10^{13} ohm-m (geometric middle)
Solid Insulation / 1 kHz					
Epoxy (silica filler) (1 kHz)	3.165×10^{-12}	3.601	0.019	[3]	ranges 3.2-4.5 / 0.8-3.0 x 10^{-2} / 3.2-4.0 / 0.8-3.0 x 10^{-2} / 3.0-3.8 / 2-4 x 10^{-2} (middles used); rho: Callister B.9 epoxy 10^{10}-10^{13} ohm-m (geometric middle)
Solid Insulation / 1 MHz					
Epoxy (silica filler) (1 MHz)	3.165×10^{-12}	3.402	0.03	[3]	ranges 3.2-4.5 / 0.8-3.0 x 10^{-2} / 3.2-4.0 / 0.8-3.0 x 10^{-2} / 3.0-3.8 / 2-4 x 10^{-2} (middles used); rho: Callister B.9 epoxy 10^{10}-10^{13} ohm-m (geometric middle)
Solid Insulation / 60 Hz					
Silicone rubber (60 Hz)	1×10^{-13}	3.651	0.0225	[3]	ranges 3.3-4.0 / 1.5-3.0 x 10^{-2} / 3.1-3.7 / 3.0-5.0 x 10^{-3} (middles used); rho: Callister B.9 silicone elastomer 10^{13} ohm-m
Solid Insulation / 1 MHz					
Silicone rubber (1 MHz)	1×10^{-13}	3.4	0.004	[3]	ranges 3.3-4.0 / 1.5-3.0 x 10^{-2} / 3.1-3.7 / 3.0-5.0 x 10^{-3} (middles used); rho: Callister B.9 silicone elastomer 10^{13} ohm-m
Insulating Liquids (60 Hz)					
Capacitor oil (60 Hz)	0	2.2	0.001	[3]	dielectric constant at 25 C, dissipation factor at 100 C
Pipe cable oil (60 Hz)	0	2.15	0.001	[3]	dielectric constant at 25 C, dissipation factor at 100 C
Self-contained cable oil (60 Hz)	0	2.3	0.001	[3]	dielectric constant at 25 C, dissipation factor at 100 C
Heavy cable oil (60 Hz)	0	2.23	0.001	[3]	dielectric constant at 25 C, dissipation factor at 100 C
Transformer oil (60 Hz)	0	2.25	0.001	[3]	dielectric constant at 25 C, dissipation factor at 100 C
Alkyl benzene (60 Hz)	0	2.1	4×10^{-4}	[3]	dielectric constant at 25 C, dissipation factor at 100 C
Polybutene pipe cable oil (60 Hz)	0	2.14	3×10^{-4}	[3]	dielectric constant at 25 C, dissipation factor at 100 C; dielectric constant given at 1 MHz
Polybutene capacitor oil (60 Hz)	0	2.22	5×10^{-4}	[3]	dielectric constant at 25 C, dissipation factor at 100 C; dielectric constant given at 1 MHz
Silicone fluid (60 Hz)	0	2.7	1.5×10^{-4}	[3]	dielectric constant at 25 C, dissipation factor at 100 C
Castor oil (60 Hz)	0	3.747	0.06	[3]	dielectric constant at 25 C, dissipation factor at 100 C

Entry	σ (S/m)	ε_r	$\tan\delta$	Src.	Note
C8F16O fluorocarbon (60 Hz)	0	1.86	5×10^{-4}	[3]	dielectric constant at 25 C, dissipation factor at 100 C; dissipation factor <0.0005 (bound used)
Circuit Board Laminates (1 MHz)					
FR-4 epoxy-glass (1 MHz)	0	4.352	0.029	[4]	ranges 4.10-4.60 / 0.028-0.030 (middles used)
Polyfunctional FR-4 (1 MHz)	0	4.352	0.029	[4]	ranges 4.10-4.60 / 0.028-0.030 (middles used)
Multifunctional FR-4 (1 MHz)	0	4.352	0.029	[4]	ranges 4.10-4.60 / 0.028-0.030 (middles used)
Polyimide-glass (1 MHz)	0	4	0.0125	[4]	ranges 3.95-4.05 / 0.01-0.015 (middles used)
BT epoxy (1 MHz)	0	3.975	0.012	[4]	ranges 3.85-4.10 / 0.011-0.013 (middles used)
Cyanate ester-glass (1 MHz)	0	3.55	0.006	[4]	ranges 3.50-3.60 / 0.005-0.007 (middles used)
PPO epoxy (1 MHz)	0	4	0.0085	[4]	ranges 3.80-4.20 / 0.008-0.009 (middles used)
PTFE-glass (1 MHz)	0	2.5	0.002	[4]	ranges 2.45-2.55 / 0.001-0.003 (middles used)
Alumina substrate (1 MHz)	0	8.9	8.5×10^{-4}	[4]	dissipation factor .0002-.0015 (middle used)
Water					
Pure water, 20 C	4.193×10^{-6}	80.07	0	[5, 7]	eps_r: P.527-6 eq. 8 (static; Debye relaxation above 17 GHz, so constant below 1 MHz); sigma: ultrapure water 23.85 MOhm-cm (Light et al.)
Pure water, 25 C	5.501×10^{-6}	78.3	0	[5, 7]	eps_r: P.527-6 eq. 8 (static; Debye relaxation above 17 GHz, so constant below 1 MHz); sigma: ultrapure water 18.18 MOhm-cm (Light et al.)
Sea water, 35 ppt, 0 C	2.904	78.6	0	[5]	P.527-6 eqs. 17 and 22-27, salinity 35 g/kg; frequency-independent below 1 MHz
Sea water, 35 ppt, 10 C	3.809	75.01	0	[5]	P.527-6 eqs. 17 and 22-27, salinity 35 g/kg; frequency-independent below 1 MHz
Sea water, 35 ppt, 20 C	4.791	71.67	0	[5]	P.527-6 eqs. 17 and 22-27, salinity 35 g/kg; frequency-independent below 1 MHz
Sea water, low salinity (ITU-R P.368)	1	80	0	[6]	Figure 1; standard ground constants for ground-wave curves
Sea water, average salinity (ITU-R P.368)	5	70	0	[6]	Figure 2; standard ground constants for ground-wave curves
Fresh water (ITU-R P.368)	0.003	80	0	[6]	Figure 3; standard ground constants for ground-wave curves
Soil and Ground					
Land, sigma 3e-2 S/m (ITU-R P.368)	0.03	40	0	[6]	Figure 4; standard ground constants for ground-wave curves
Wet ground (ITU-R P.368)	0.01	30	0	[6]	Figure 5; standard ground constants for ground-wave curves
Land, sigma 3e-3 S/m (ITU-R P.368)	0.003	22	0	[6]	Figure 6; standard ground constants for ground-wave curves
Medium dry ground (ITU-R P.368)	0.001	15	0	[6]	Figure 7; standard ground constants for ground-wave curves

Entry	σ (S/m)	ε_r	$\tan\delta$	Src.	Note
Dry ground (ITU-R P.368)	3×10^{-4}	7	0	[6]	Figure 8; standard ground constants for ground-wave curves
Very dry ground (ITU-R P.368)	1×10^{-4}	3	0	[6]	Figure 9; standard ground constants for ground-wave curves
Fresh water ice, -1 C (ITU-R P.368)	3×10^{-5}	3	0	[6]	Figure 10; standard ground constants for ground-wave curves
Fresh water ice, -10 C (ITU-R P.368)	1×10^{-5}	3	0	[6]	Figure 11; standard ground constants for ground-wave curves

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